Methods: Event-Based Time-Sampling Evaluation

For each site and day, timestamps were discretized into fixed-width time bins (bin widths = 5, 10, 15, 30, and 60 minutes). Only bins containing at least one detection event were considered *available bins*. The full set of available bins for a day constituted the ground-truth representation of the bird community for that day.

Candidate time-sampling strategies were evaluated by exhaustively enumerating time-sample subsets of bins per day and comparing the resulting sampled community to the full-day ground truth. Two classes of time-sampling strategies were tested: (a) fixed-time sampling, where the same clock-time bin was selected each day when available, and (b) random-time sampling, where bins were selected uniformly at random from the available bins for that day. Sampling effort was quantified as the number of bins sampled per day. For each site, strategies were evaluated exhaustively across all feasible effort levels for each bin width (i.e., from one bin per day up to the maximum number of event-containing bins observed in a day).

Accuracy was quantified in two ways (see Glossary of terms below for definitions and equations). First, relative to species diversity: Jaccard similarity and a similarity transform of Bray–Curtis were used to quantify differences between the full-day community and the sampled community. Second, relative to total count of bird detections: the mean absolute error (MAE) between sampled and true daily totals was computed, and then converted to mean absolute percentage error (MAPE). All metrics were computed at the daily level and subsequently averaged across days within site.

The result of running the algorithm was 33,526 comparisons of time-sampling strategies relative to the collected data.

Results

Below are six plots representing the time-sampling results for each of three locations. Each row in the figures represents a different accuracy metric with the top two rows focusing on accuracy of capturing species diversity and the bottom two rows focusing vist count accuracy. Each column represents one of the time-sample windows tested; that is 5-, 10-, 15-, 30-, or 60-minutes of continuous observation. Black markers (Figure 1) or density estimation (Figure 2) are the results from sampling at a fixed-time every day (i.e., the same time tested every day). Red markers or density estimation are the results from sampling at a random-time every day.

The first figure for each location is a scatterplot showing the results from one of the 33,526 time-sample tests conducted. Because these are a bit difficult to see, the second figure was created which is the exaact same data as in Figure 1, but converted from a scatterplot to a kernel density estimation (KDE) plot.

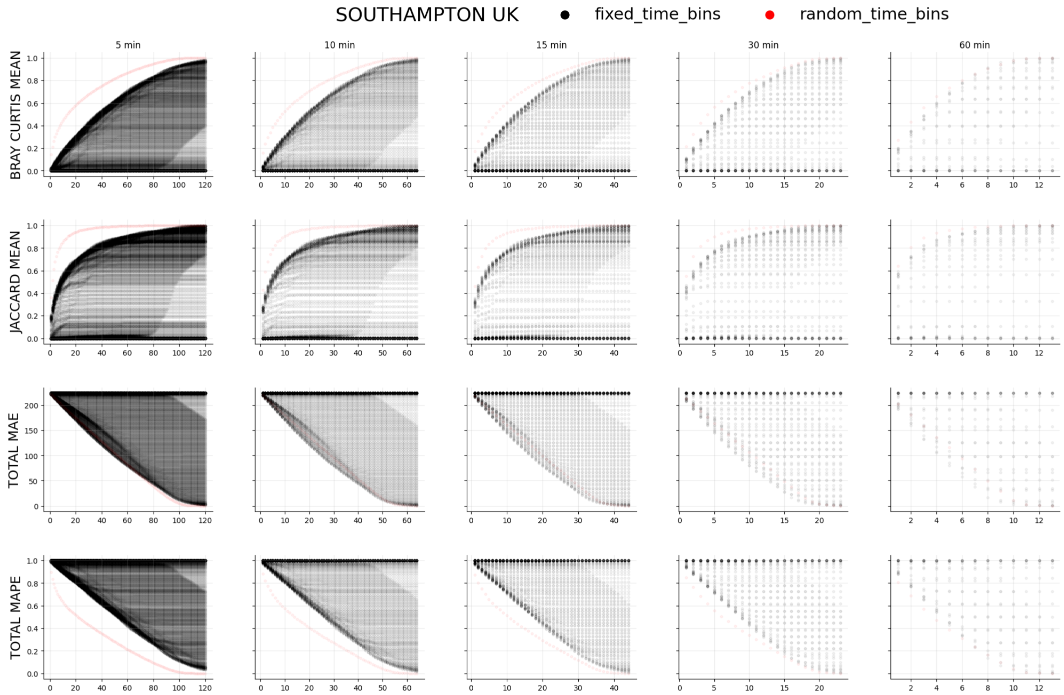

Figure 1: Effort–fidelity scatterplot for Southampton, UK.

Figure 2: Effort–fidelity kernel density estimation (KDE) plot for Southampton, UK.

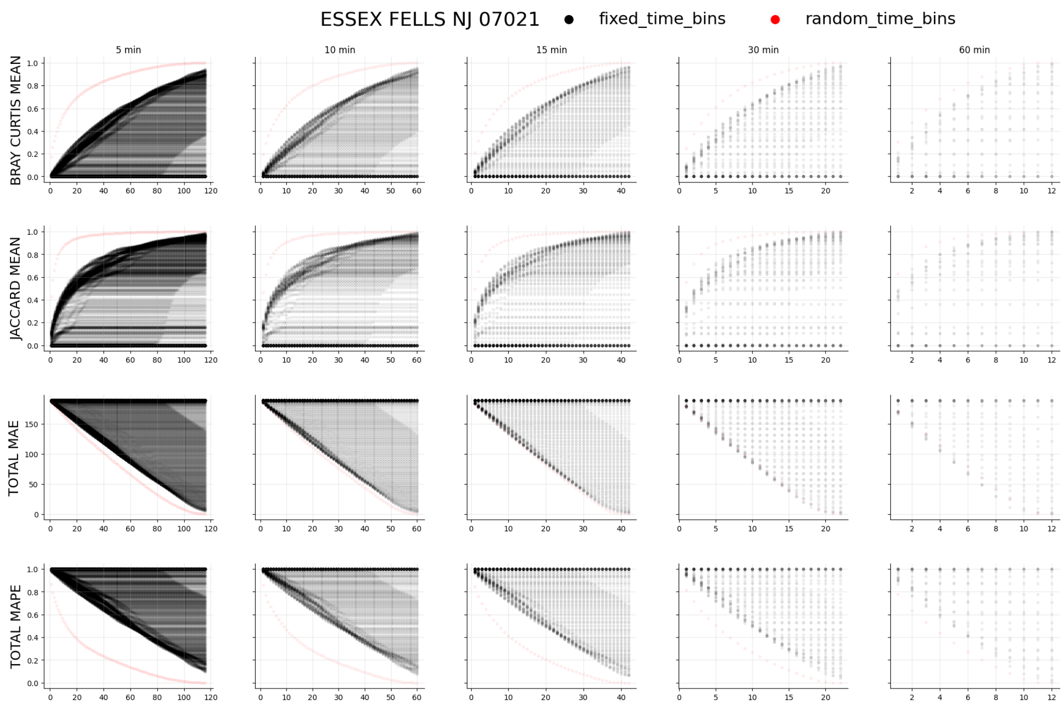

Figure 3: Effort–fidelity scatterplot for Essex Fells, NJ.

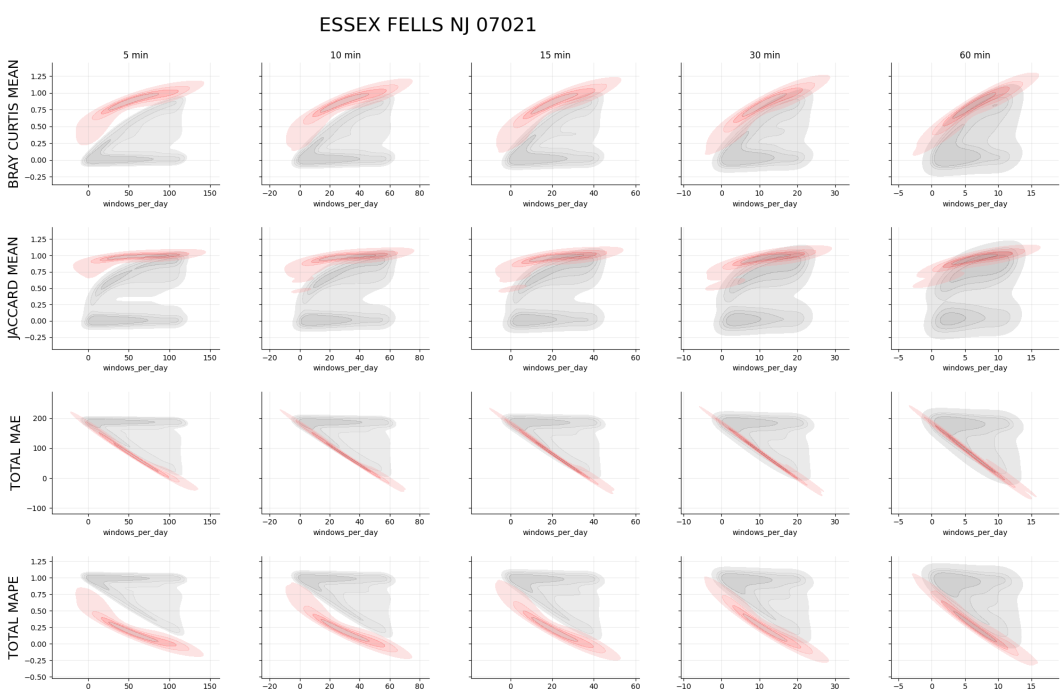

Figure 4: Effort–fidelity kernel density estimation (KDE) plot for Essex Fells, NJ.

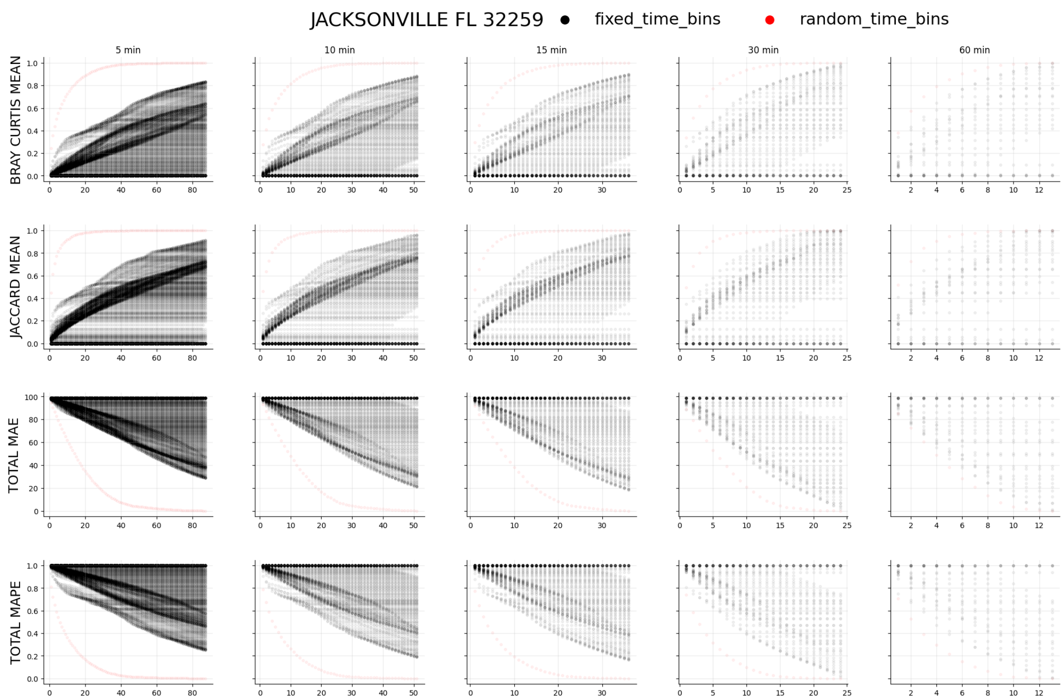

Figure 5: Effort–fidelity scatterplot for Jacksonville, FL.

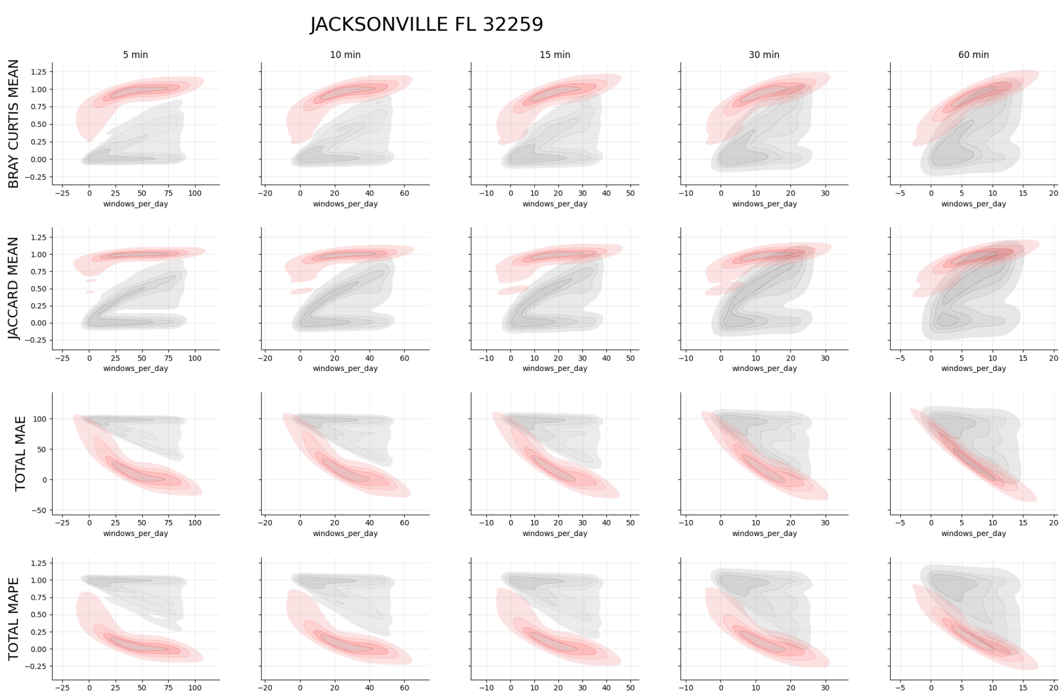

Figure 6: Effort–fidelity kernel density estimation (KDE) plot for Jacksonville, FL.

Glossary of Evaluation Metrics

days Number of unique days included in the evaluation.

$$\text{days} = |\{\text{Date}\}|\,,$$

where each Date corresponds to one calendar day of observations.

bins_available_mean Mean number of event-containing time bins available per day.

$$\frac{1}{D}\sum_{d=1}^{D} |\{\text{bins with } \geq 1 \text{ event on day } d\}|\,,$$

where D is the total number of evaluated days.

bins_sampled_mean Mean number of bins sampled per day under a given strategy.

$$\frac{1}{D}\sum_{d=1}^{D} |\{\text{bins sampled on day } d\}|\,.$$

sampling_fraction_mean Mean proportion of available bins sampled per day.

$$\frac{\text{bins\_sampled\_mean}}{\text{bins\_available\_mean}}.$$

total_MAE Mean absolute error in total daily bird detections.

$$\frac{1}{D}\sum_{d=1}^{D} \left|N_d^{\text{sample}} - N_d^{\text{true}}\right|,$$

where N_d^{true} is the total number of detections observed on day d and N_d^{sample} is the number recovered from sampled bins.

total_MAPE Mean absolute percentage error in total daily detections.

$$\frac{1}{D}\sum_{d=1}^{D} \frac{\left|N_d^{\text{sample}} - N_d^{\text{true}}\right|}{N_d^{\text{true}}}.$$

richness_MAE Mean absolute error in daily species richness.

$$\frac{1}{D}\sum_{d=1}^{D} \left|S_d^{\text{sample}} - S_d^{\text{true}}\right|,$$

where S_d^{true} and S_d^{sample} denote the number of unique species observed in the full day and sampled bins, respectively.

jaccard_mean Mean Jaccard similarity of species presence between sampled and true communities.

$$\frac{|A \cap B|}{|A \cup B|},$$

where A is the set of species detected in sampled bins and B is the set detected in the full day.

`bray_curtis_mean` Mean Bray–Curtis similarity of species abundance distributions.

$$1 - \frac{\sum_i |x_i - y_i|}{\sum_i (x_i + y_i)},$$

where x_i is the total daily count of species i in the full dataset and y_i is the corresponding count recovered from sampled bins.

`site` Unique identifier for a data-collection location.

`strategy` Time-sampling rule applied (e.g., fixed-time or random-time sampling).

`start_minute` Clock-time anchor (minutes since midnight) used for fixed-time sampling strategies.